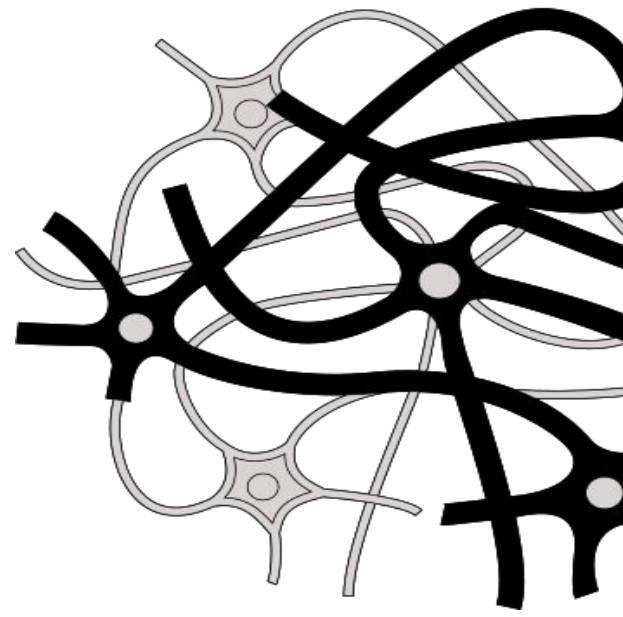
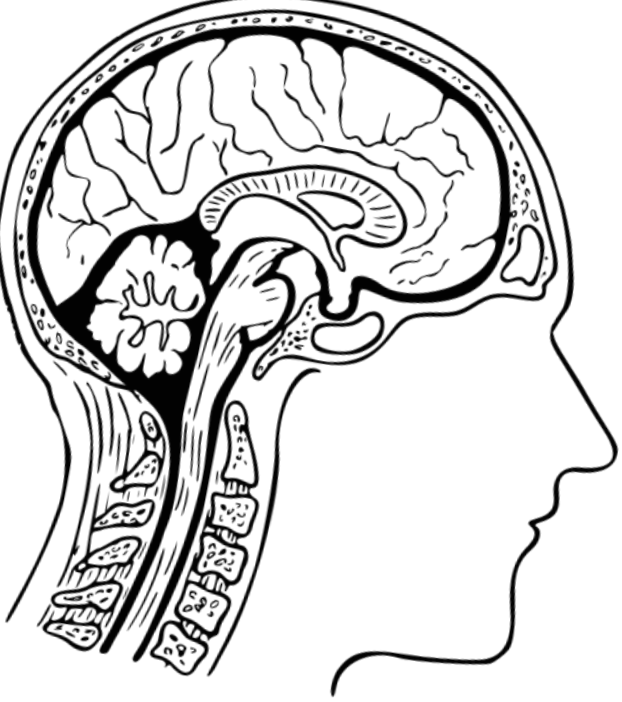
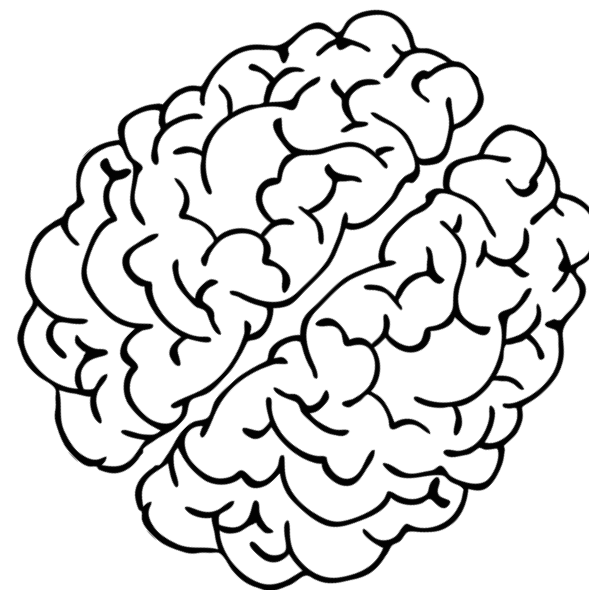
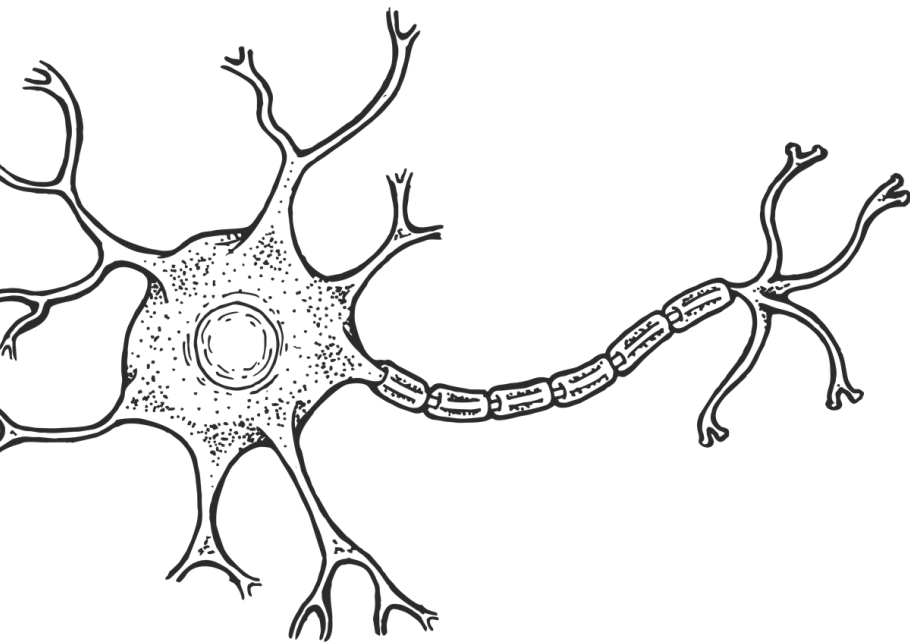
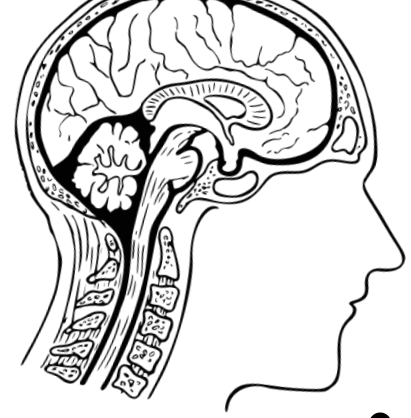




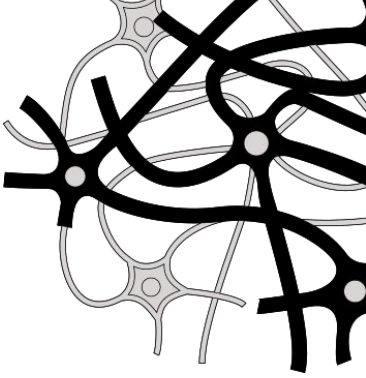
Session 1: Orientation + Neurons & the Electrical Brain

presented by Lynn

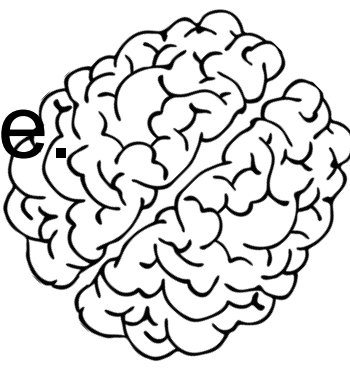
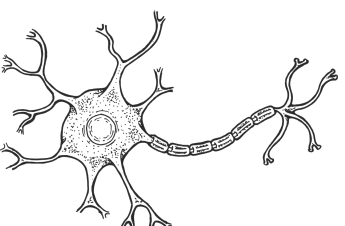


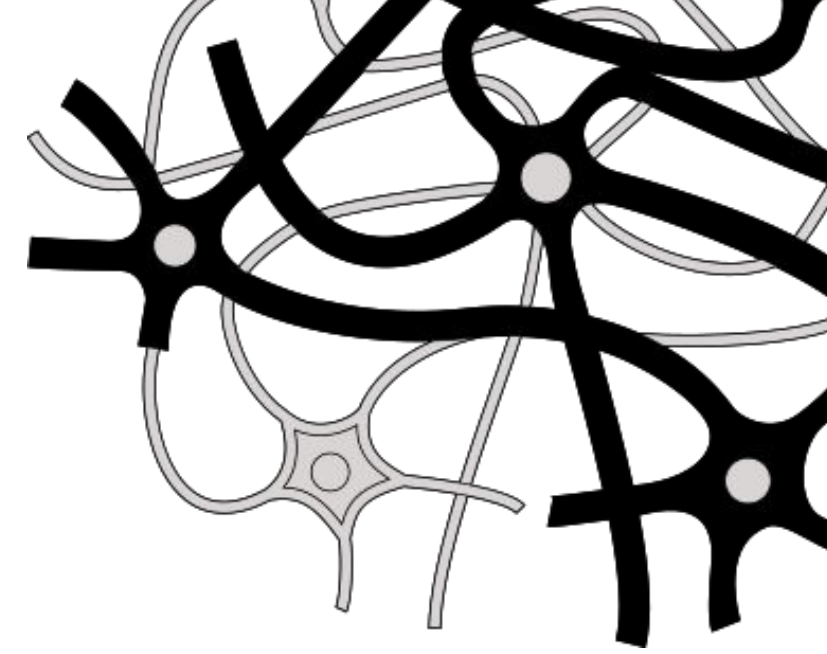
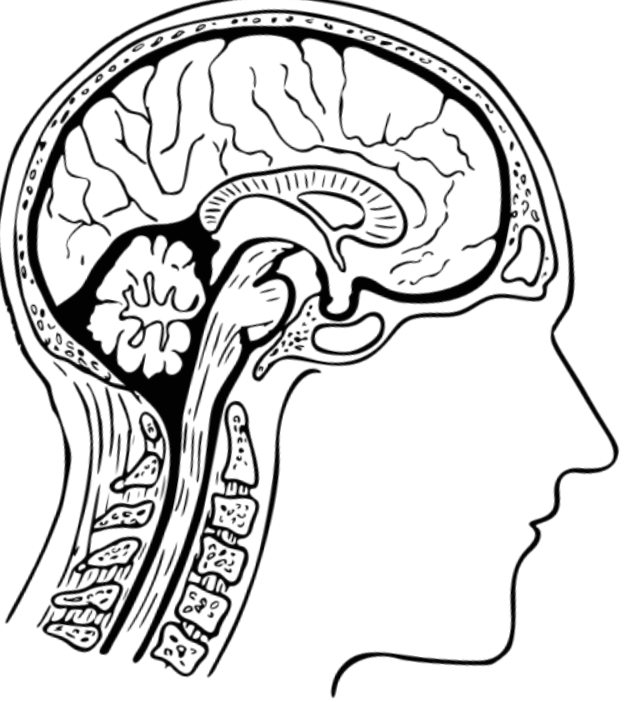


ABDN EEG Fellowship: Schedule & Infrastructure

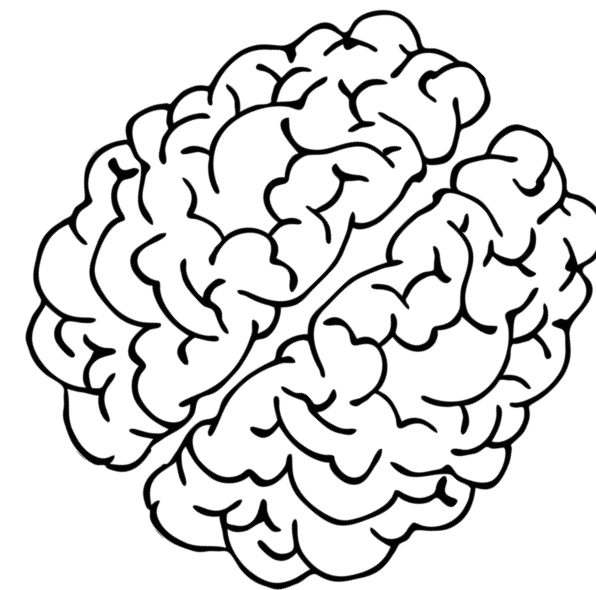
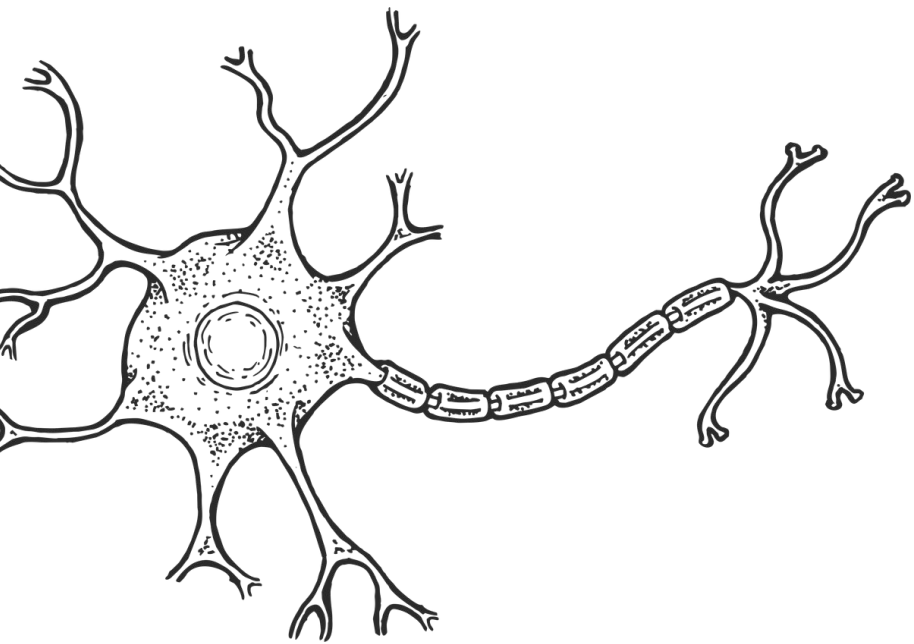


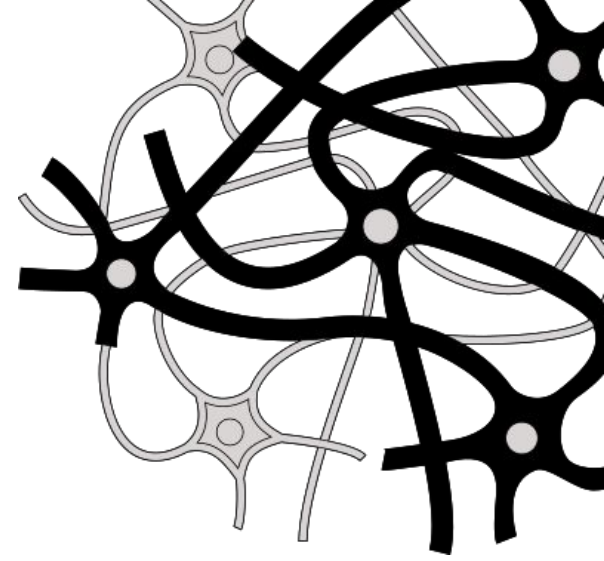
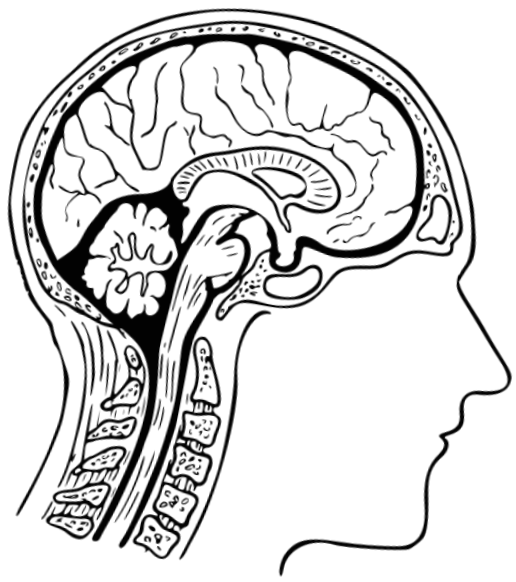
- [EEG Syllabus](#)
- Timeline
 - Fellowship will run from July to November
- Schedule and Office hours
 - Core sessions: 2 sessions per week(60-90 min each)
 - TA support: 4 teaching assistants
 - Office hours: 8 total hours per week(2 hours/week per TA)
- Technical Stack
 - Compute: Google Colab(Requires active Google account)
 - Processing: MNE-Python
 - Version Control: Github
- Verify Colab access, bookmark the GitHub repo, access shared Google Drive.





Module 0: The Brain





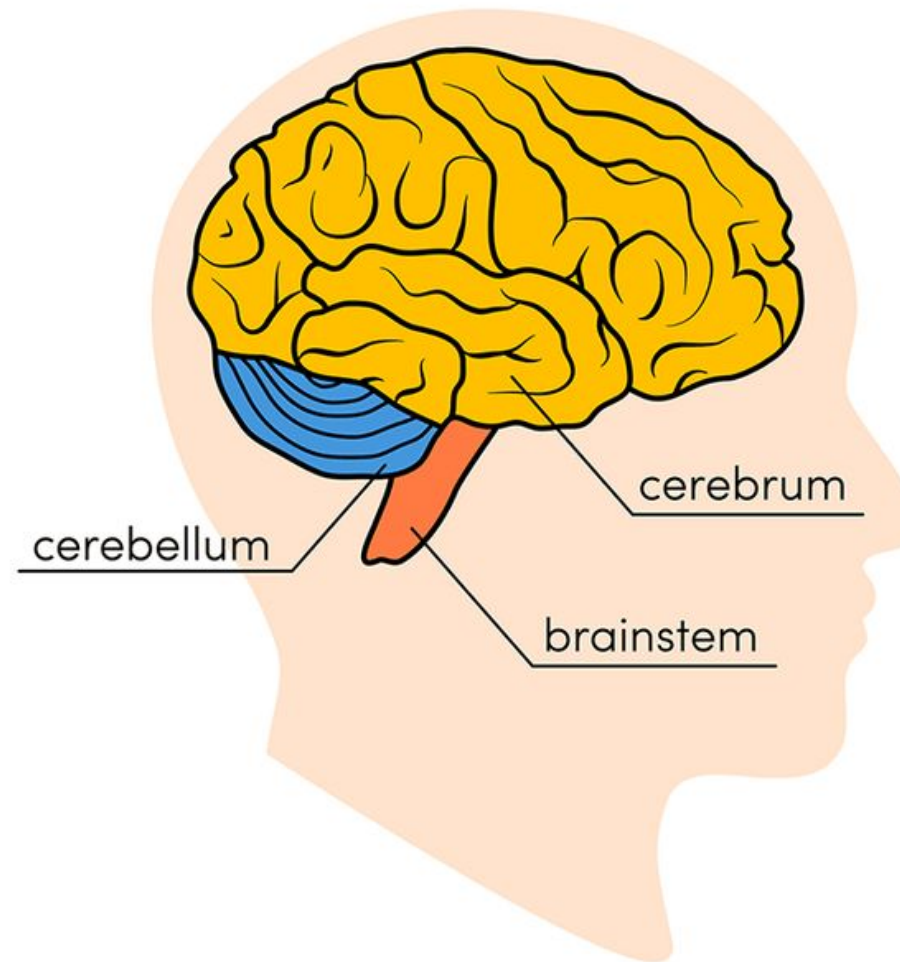
Three Major Parts

Cerebrum

The largest part. It handles higher functions like interpreting touch, vision, hearing, as well as reasoning and learning.

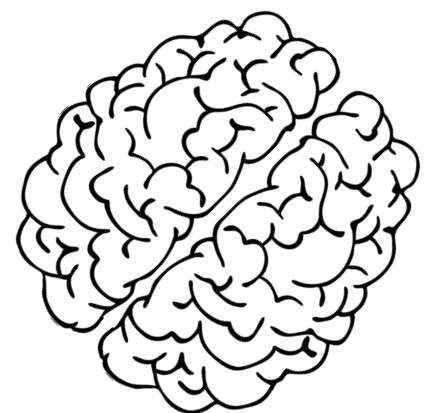
Cerebellum

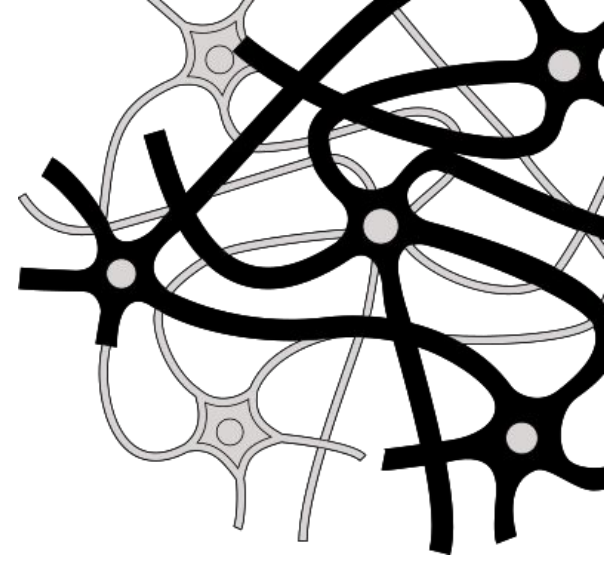
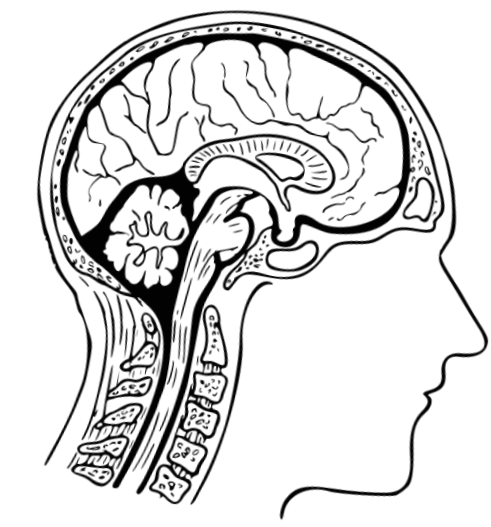
Located under the cerebrum. Its primary job is coordination, muscle movement, and maintaining balance.



Brainstem

The "relay center." It manages automatic functions like breathing, heart rate, and sleep cycles.

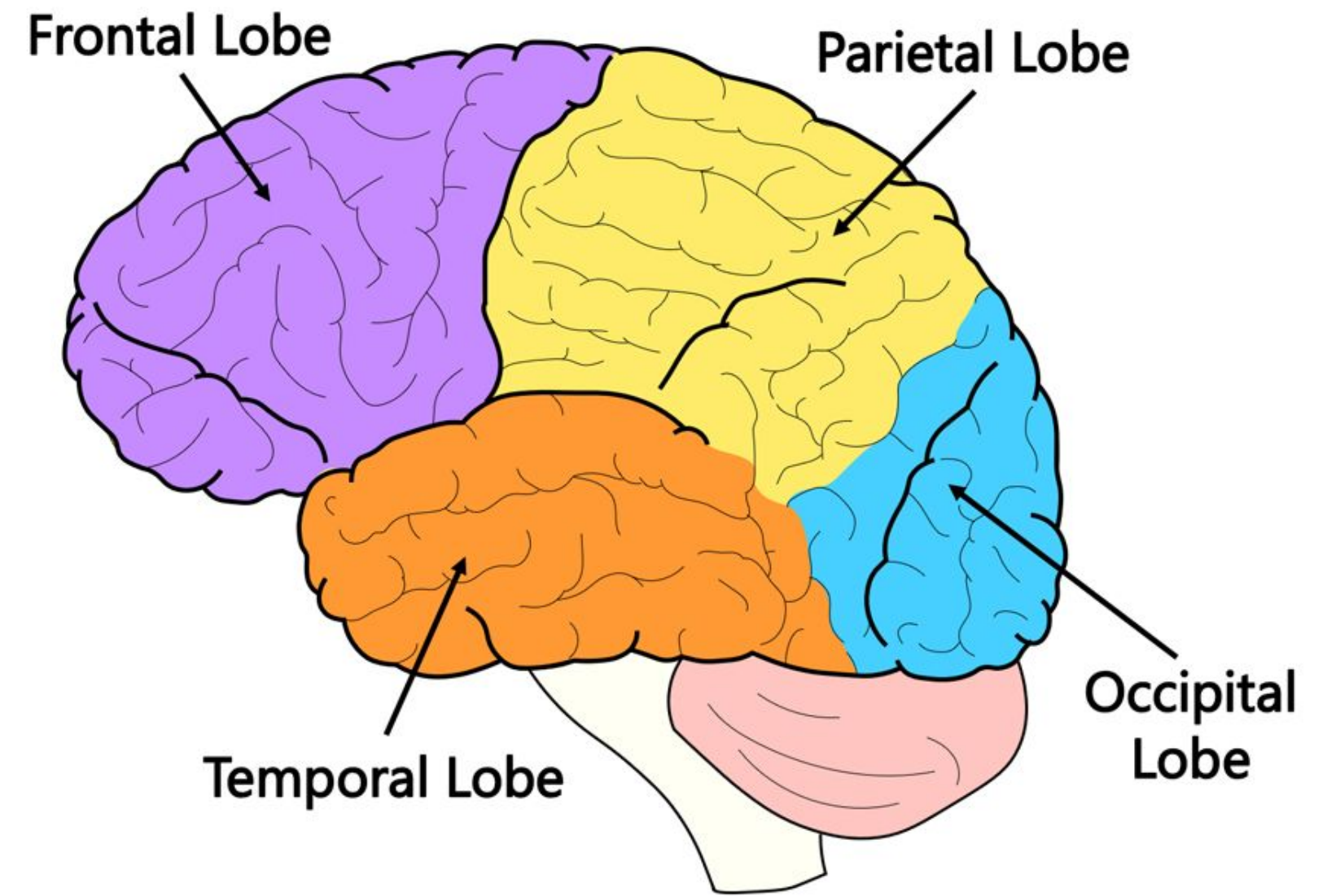
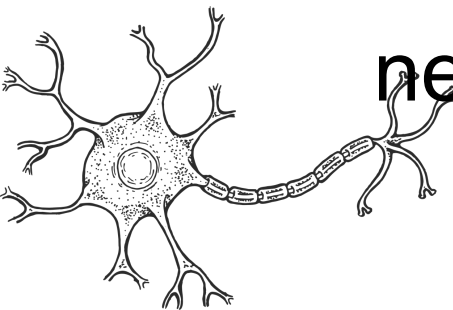


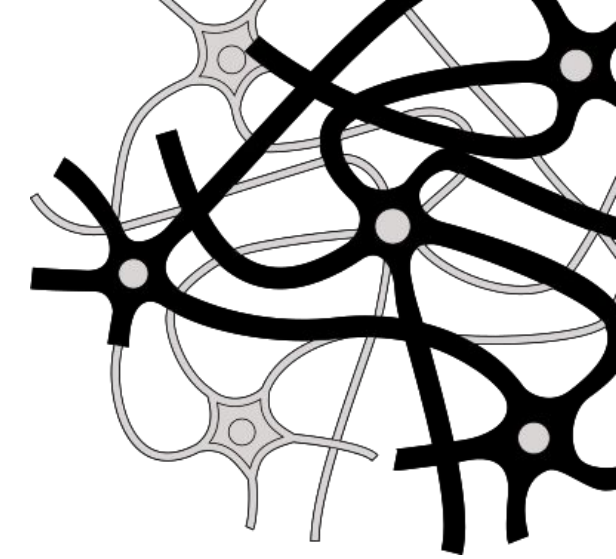
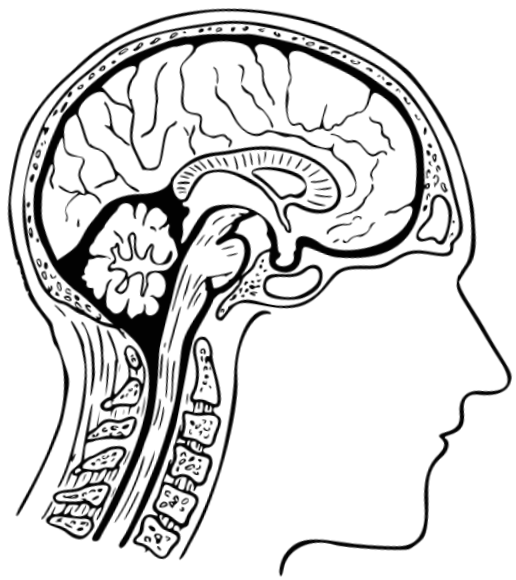


Four Lobes of the Cortex

- **Frontal:** Planning, reasoning, movement
- **Parietal:** Sensory integration(Touch, Pressure)
- **Temporal:** Hearing, Language, Memory
- **Occipital:** Visual processing centre

Each lobe has a specialised role, but all communicate constantly via massive electrical networks.

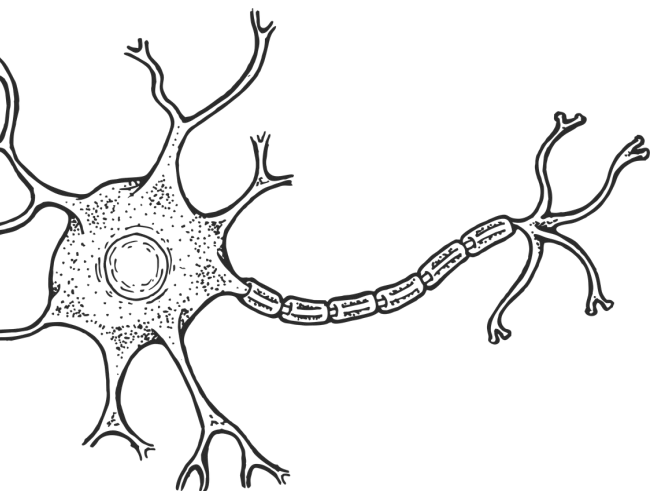




Gray vs White

Gray Matter

- Consists of neuronal cell bodies. This is where the actual processing and "thinking" happens.
- It forms the outer layer (Cortex) of the brain.



Grey Matter:

Basal ganglia

Cerebral cortex

White Matter:

Corpus callosum

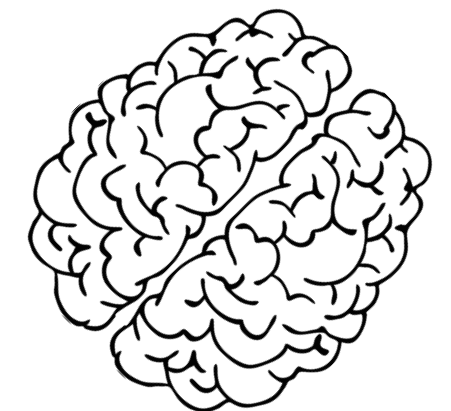
Internal capsule

Corona radiata

Coronal section

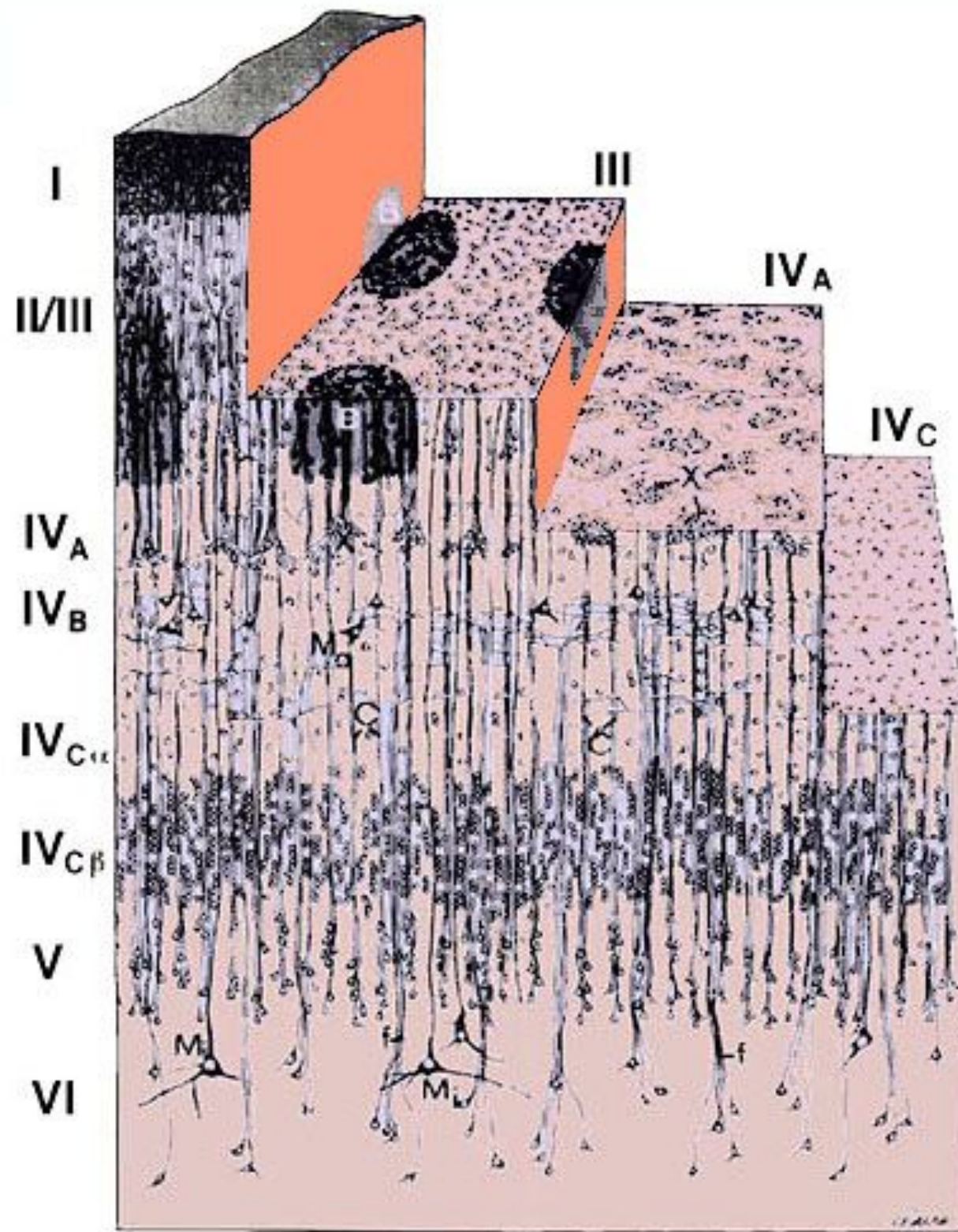
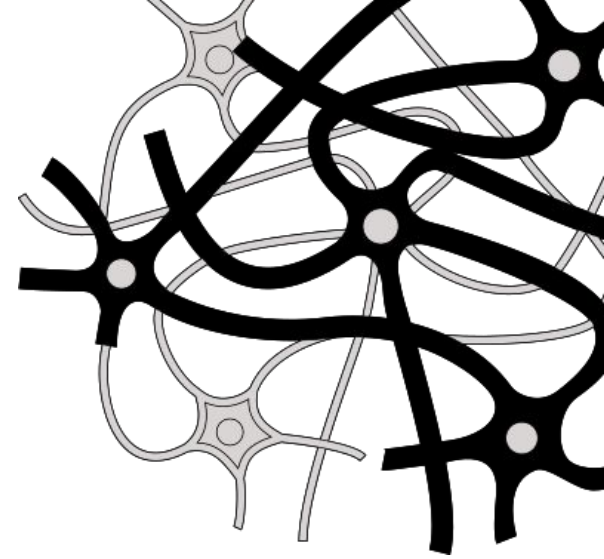
White Matter

- Made of myelinated axons (long relays). It connects different gray matter regions.
- It is white because of the fat-rich "insulation" that helps signals travel fast.





Bridge to EEG

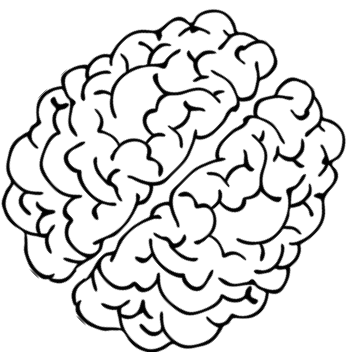


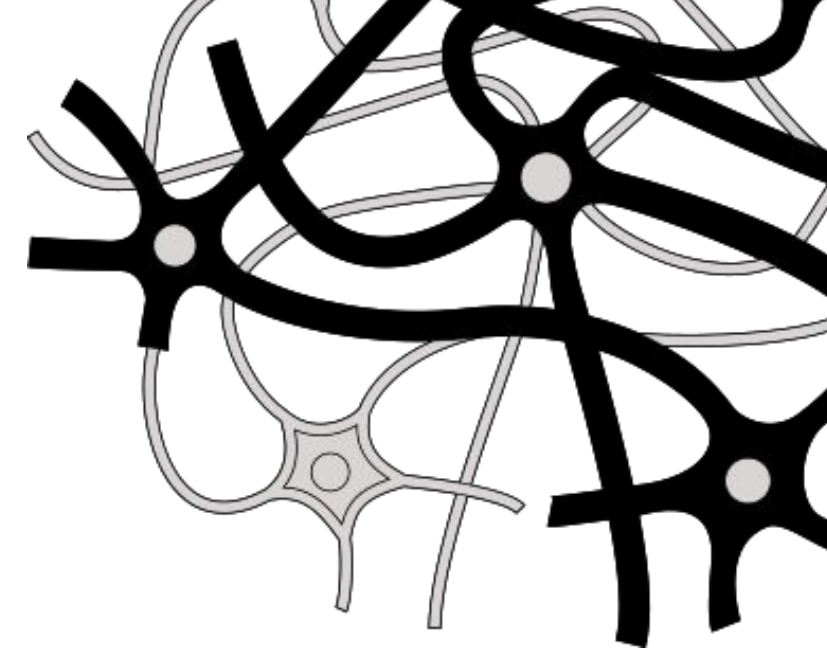
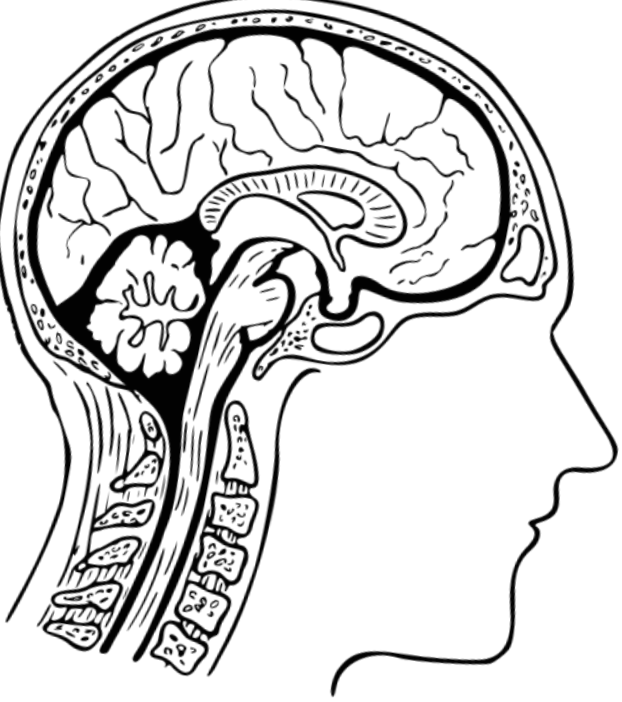
Where does the signal start?

EEG primarily measures activity from the Cerebral Cortex (the outer gray matter shell).

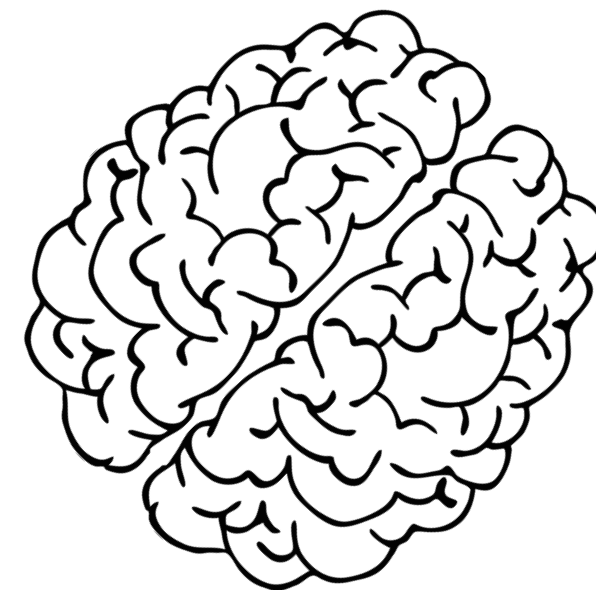
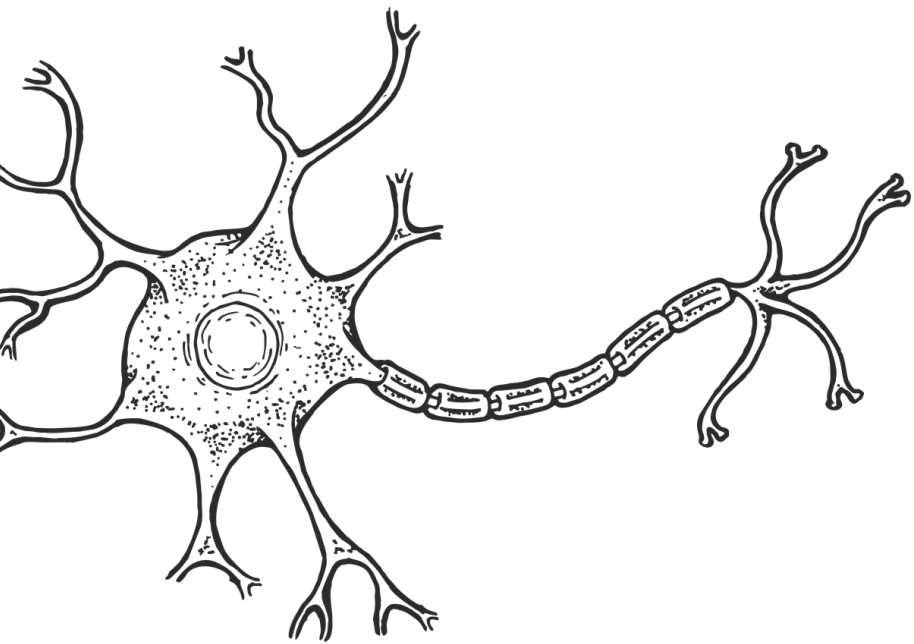
- **Structure:** The cortex is packed with "Pyramidal Cells."
- **Alignment:** These cells are perfectly aligned in parallel rows.
- **Summation:** This alignment allows their tiny electrical charges to "add up" and become strong enough to reach the scalp.

Without this specific anatomical alignment, EEG wouldn't be possible.

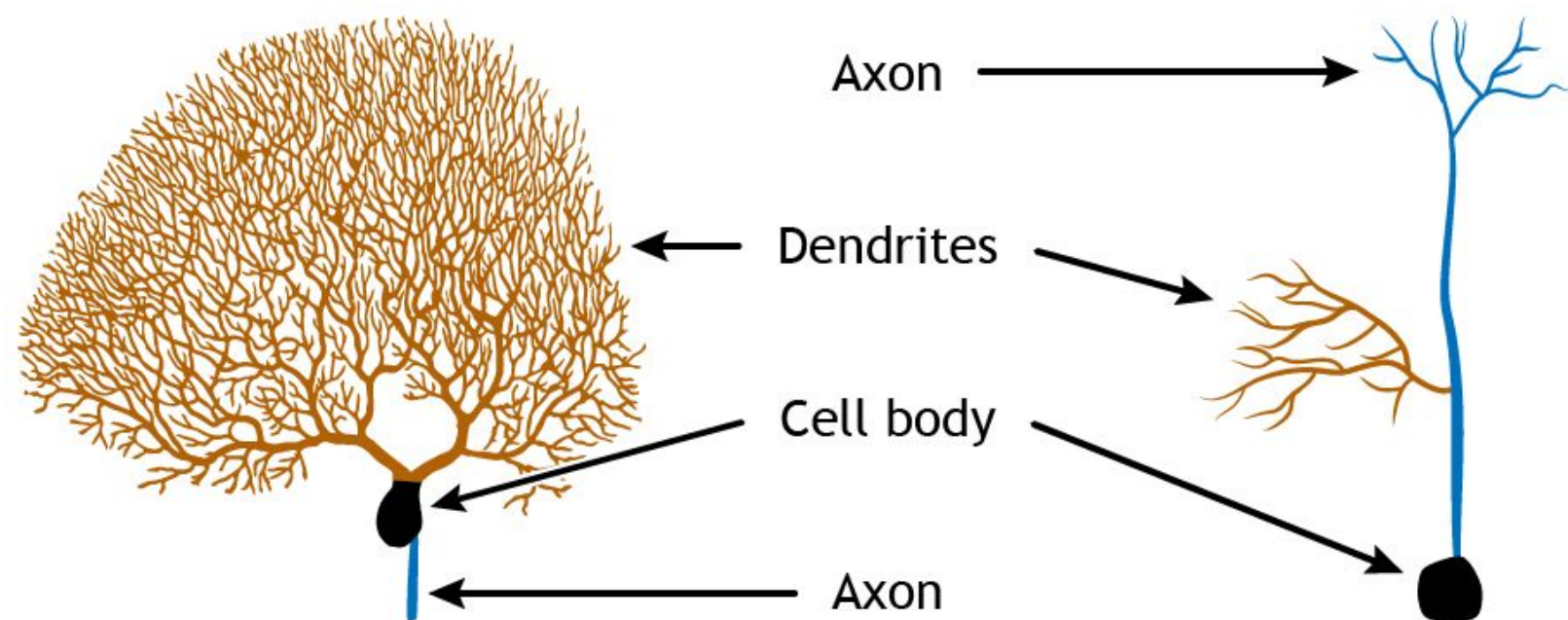
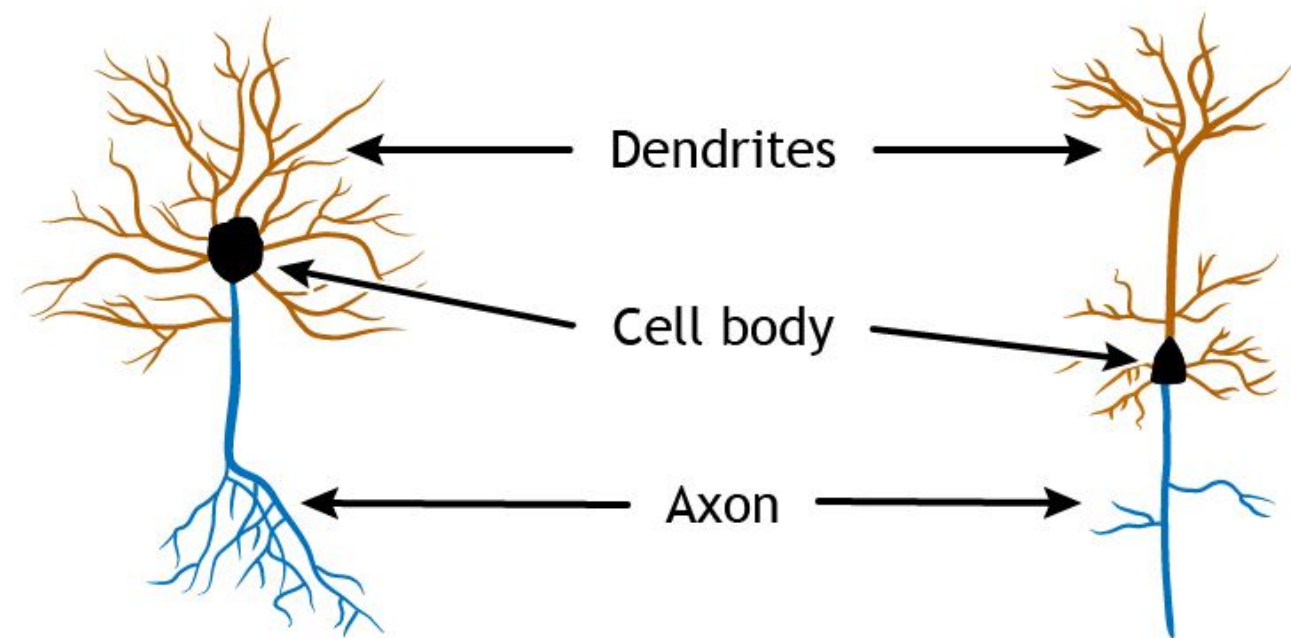
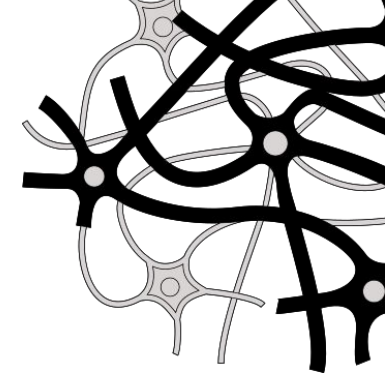




From Biology to Code

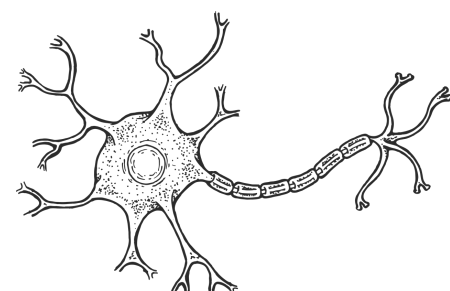


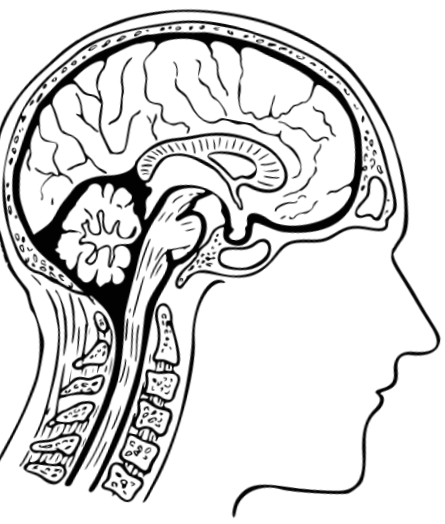
The Computational Neuron



- >> Dendritic Tree: The Input Matrix (Weighted Ligand-gated channels).
- >> Soma (Cell Body): Spatial & Temporal Integrator
- >> Axon Hillock: The Activation Function(Binary Threshold Trigger).
- >> Synapse: The Output Vector for chemical signaling

Building programmatic pipelines to extract reproducible biomarkers from cortical data.

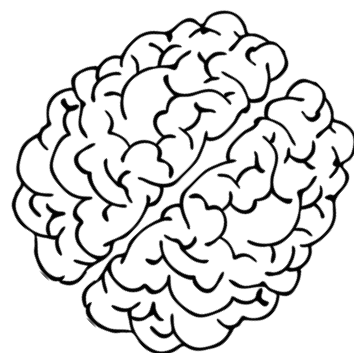
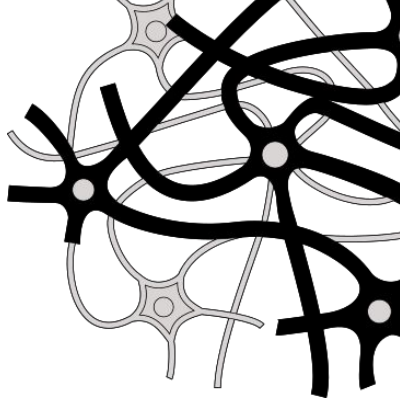
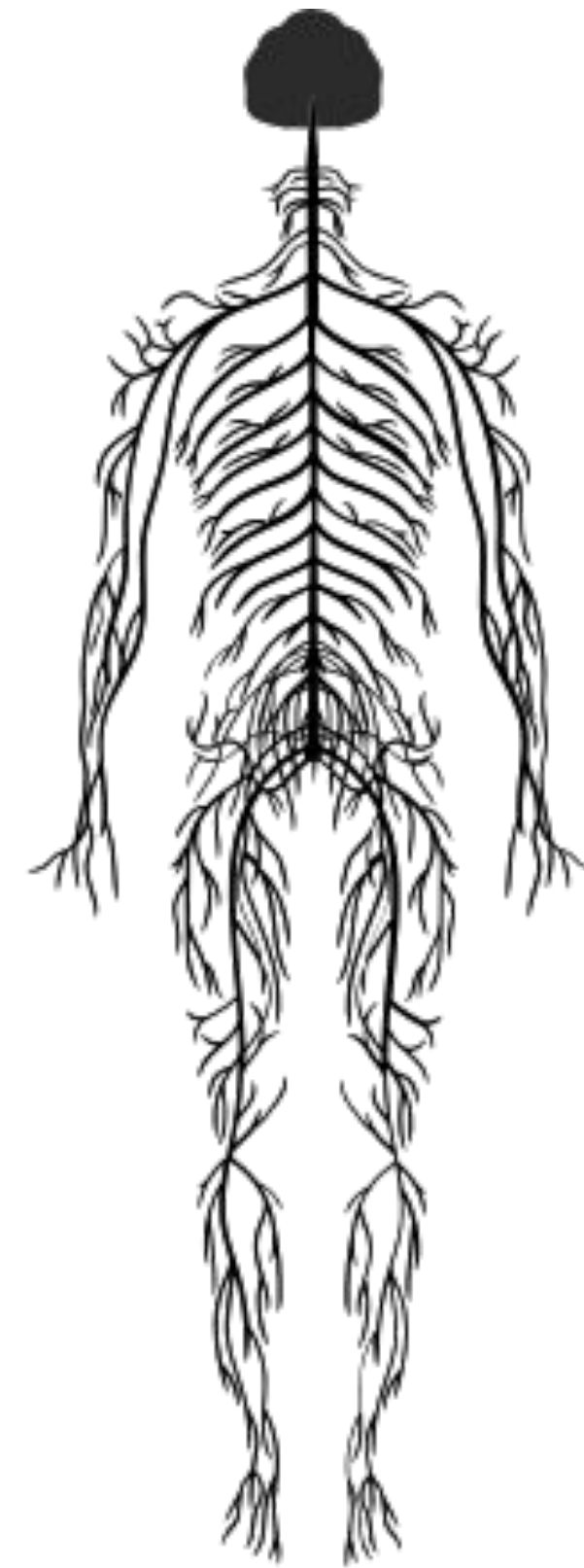
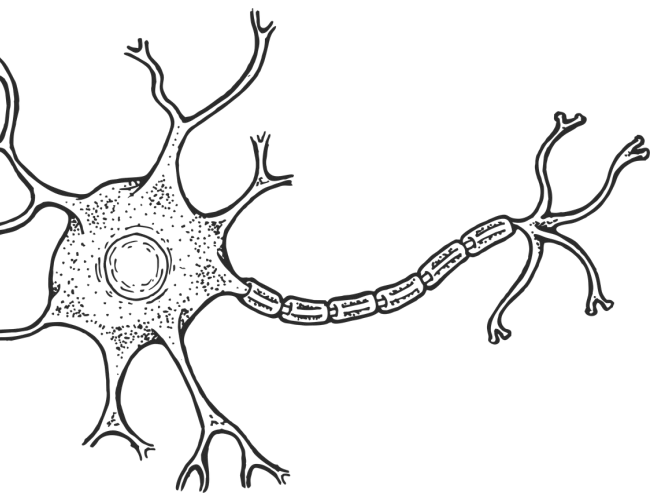




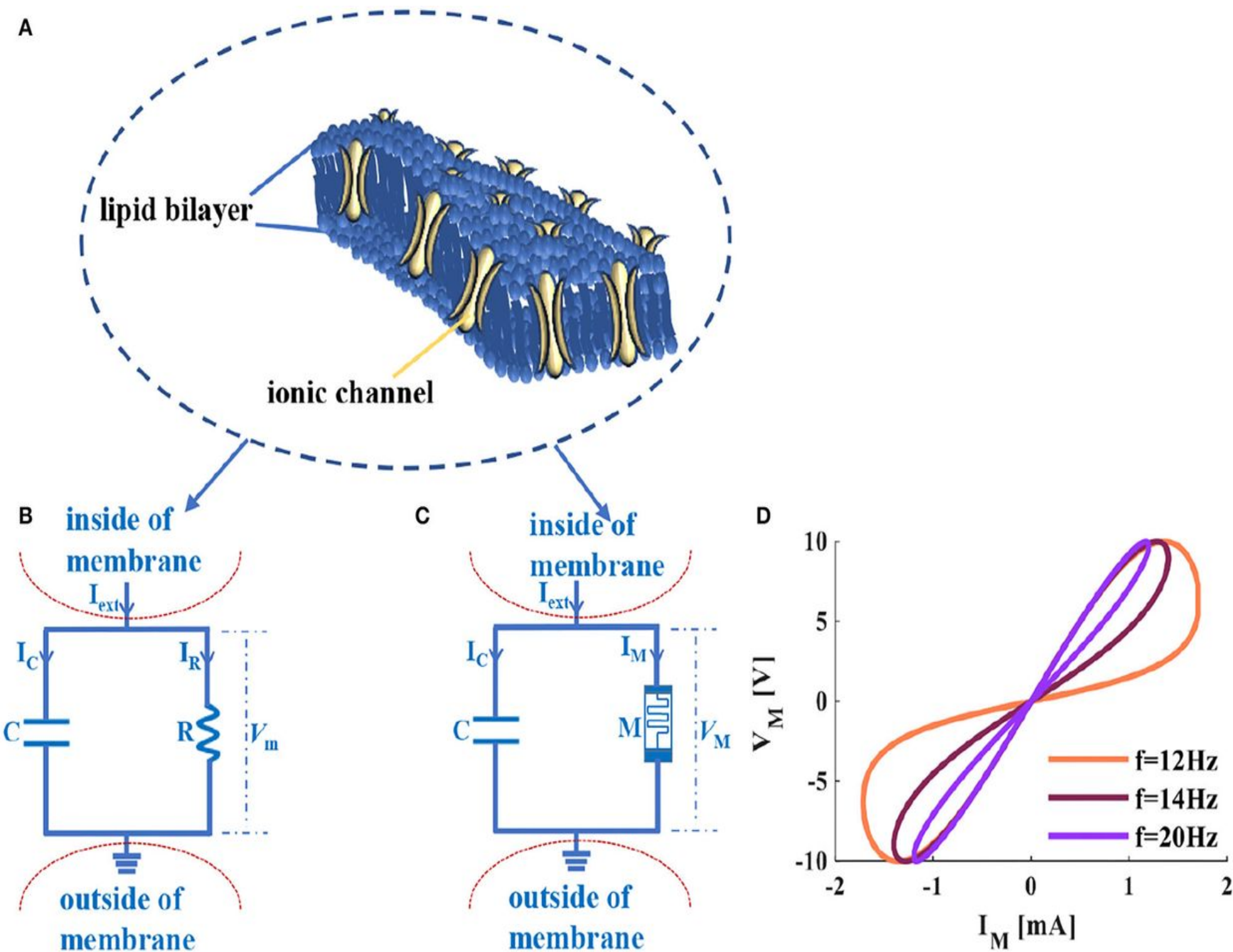
Membrane Biophysics: The RC Circuit

BIOLOGICAL COMPONENTS

- » Lipid Bilayer: Insulator/Capacitor
- » Ion Channels: Variable Resistors
- » Gradients: Batteries (Na^+/K^+)

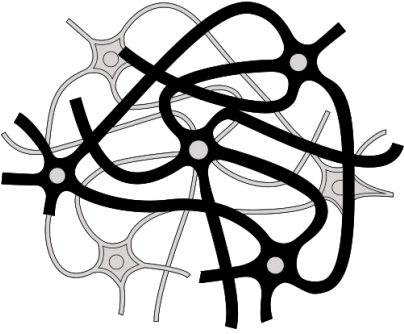
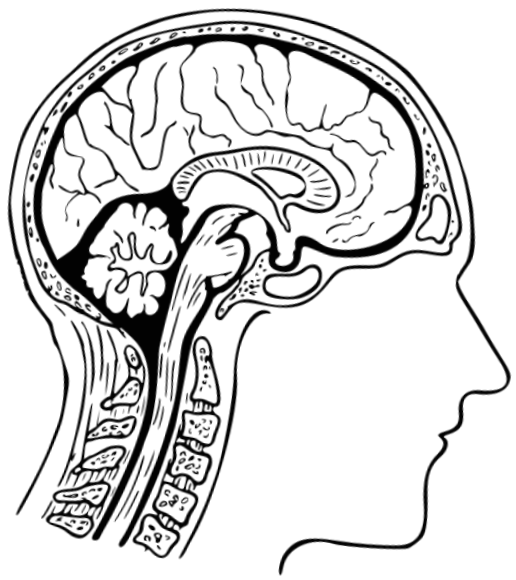


Membrane Biophysics: The RC Circuit



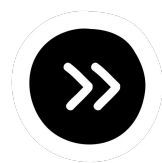
Biological Component	Electrical Equivalent	Function
Lipid Bilayer	Capacitor (C)	Insulates and stores charge
Ion Channels	Variable Resistors (R)	Controls current flow
Na ⁺ / K ⁺ Gradients	Batteries (V)	Provides the driving force

Resting State: Maintained at -70 mV. The resting potential is an active, energy-consuming baseline maintained by ionic pumps.



Action vs Post-synaptic Potentials

WHY EEG MEASURES PSPs:



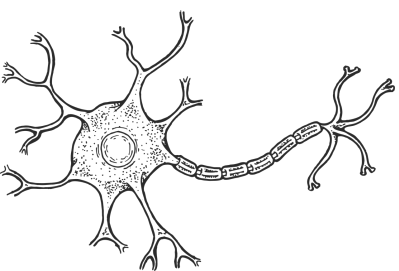
Action Potentials: Too fast ($\approx 1\text{ms}$) and asynchronous; signals cancel out at the scalp.



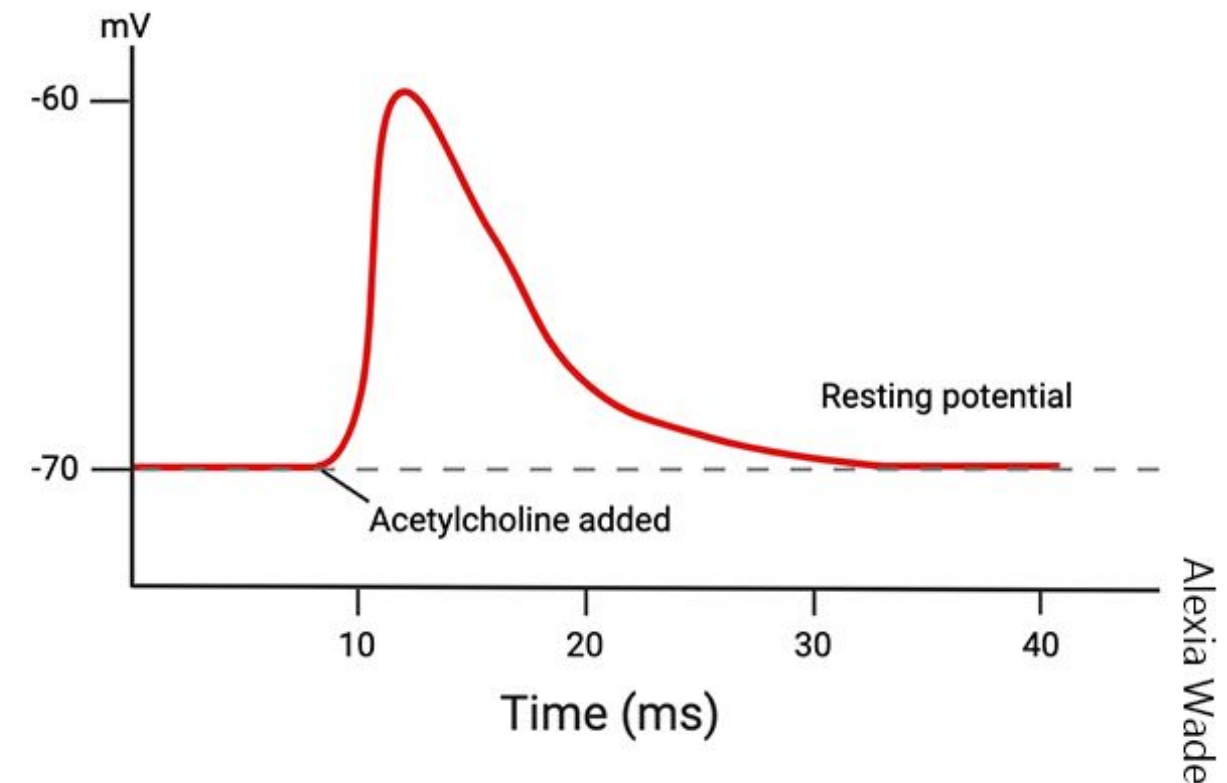
PSPs: Slower duration (10-100ms) allows for massive temporal summation.



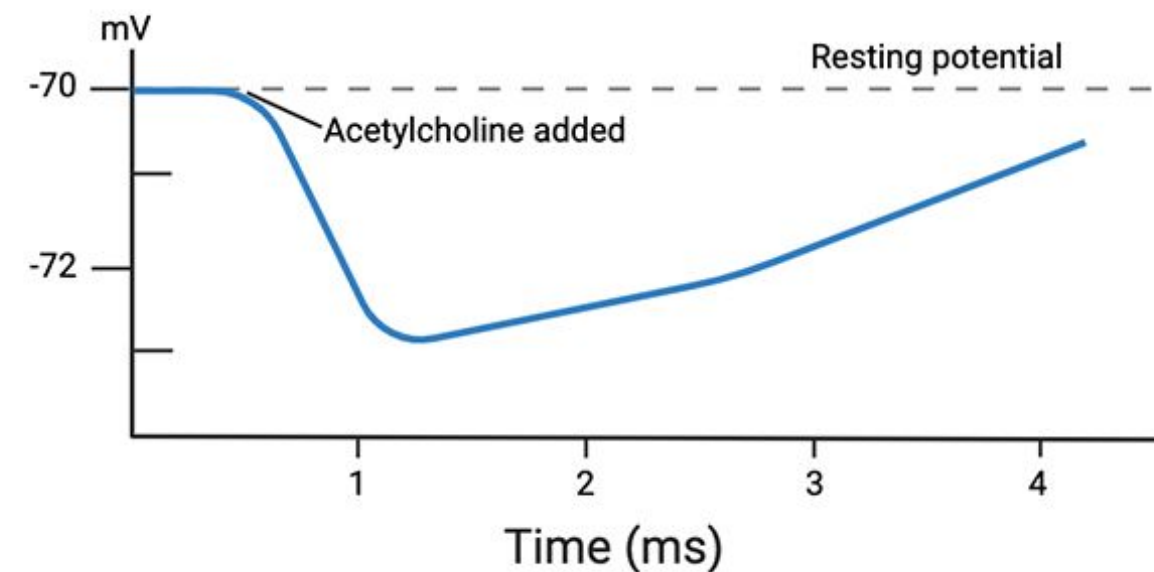
Synchrony: EEG requires thousands of neurons to fire synchronously.

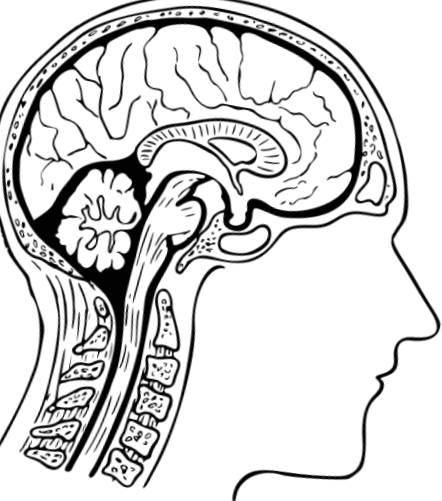


(a) Excitatory Synapse

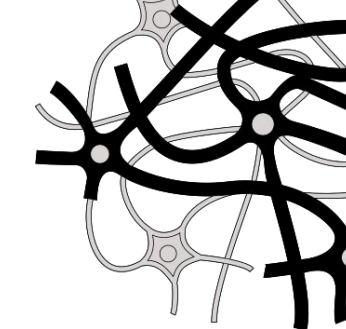


(b) Inhibitory synapse





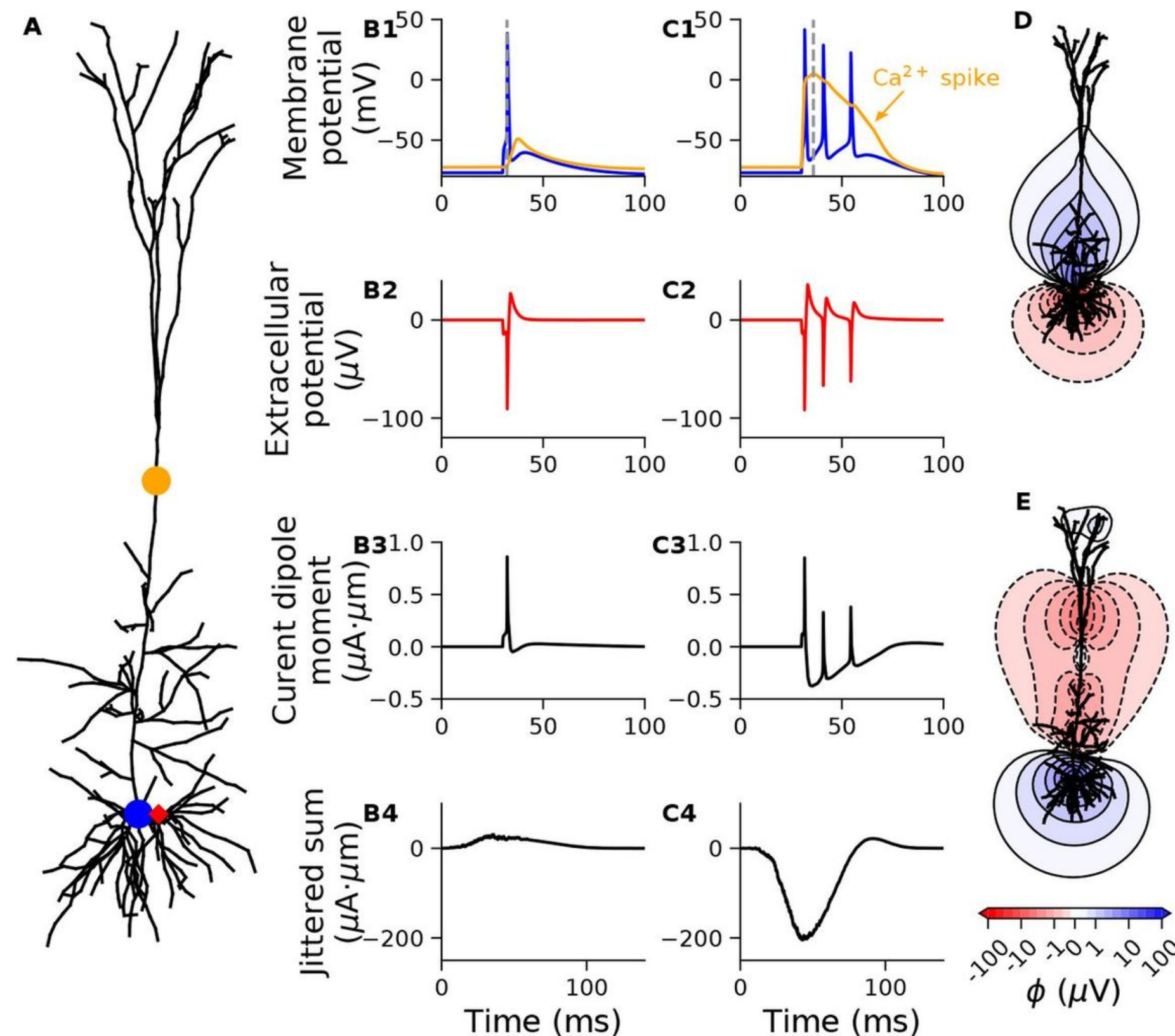
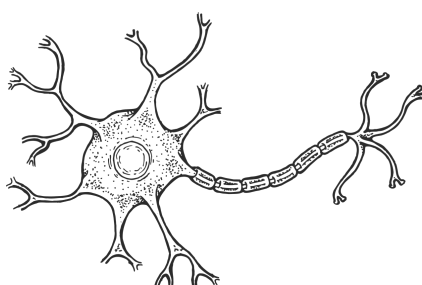
Generating the Macroscopic Signal

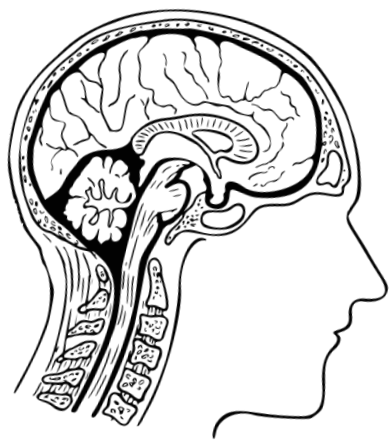


OPEN FIELD DYNAMICS

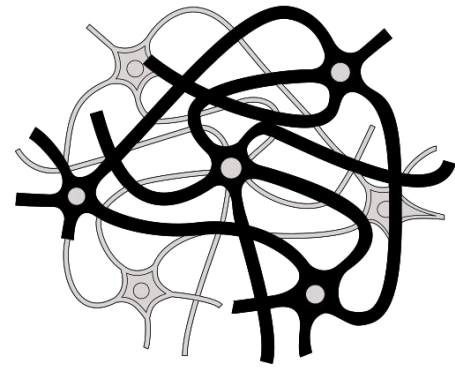
Cortical Pyramidal Neurons (Layers III & V) are perfectly aligned in parallel arrays, perpendicular to the skull.

Simultaneous excitatory input creates a massive separation of charge—a biological battery strong enough to project to the scalp.





Spatial Smearing & The Inverse Problem



The Medium

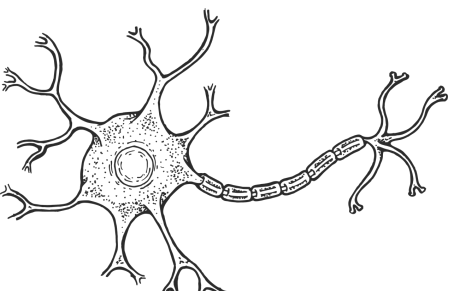
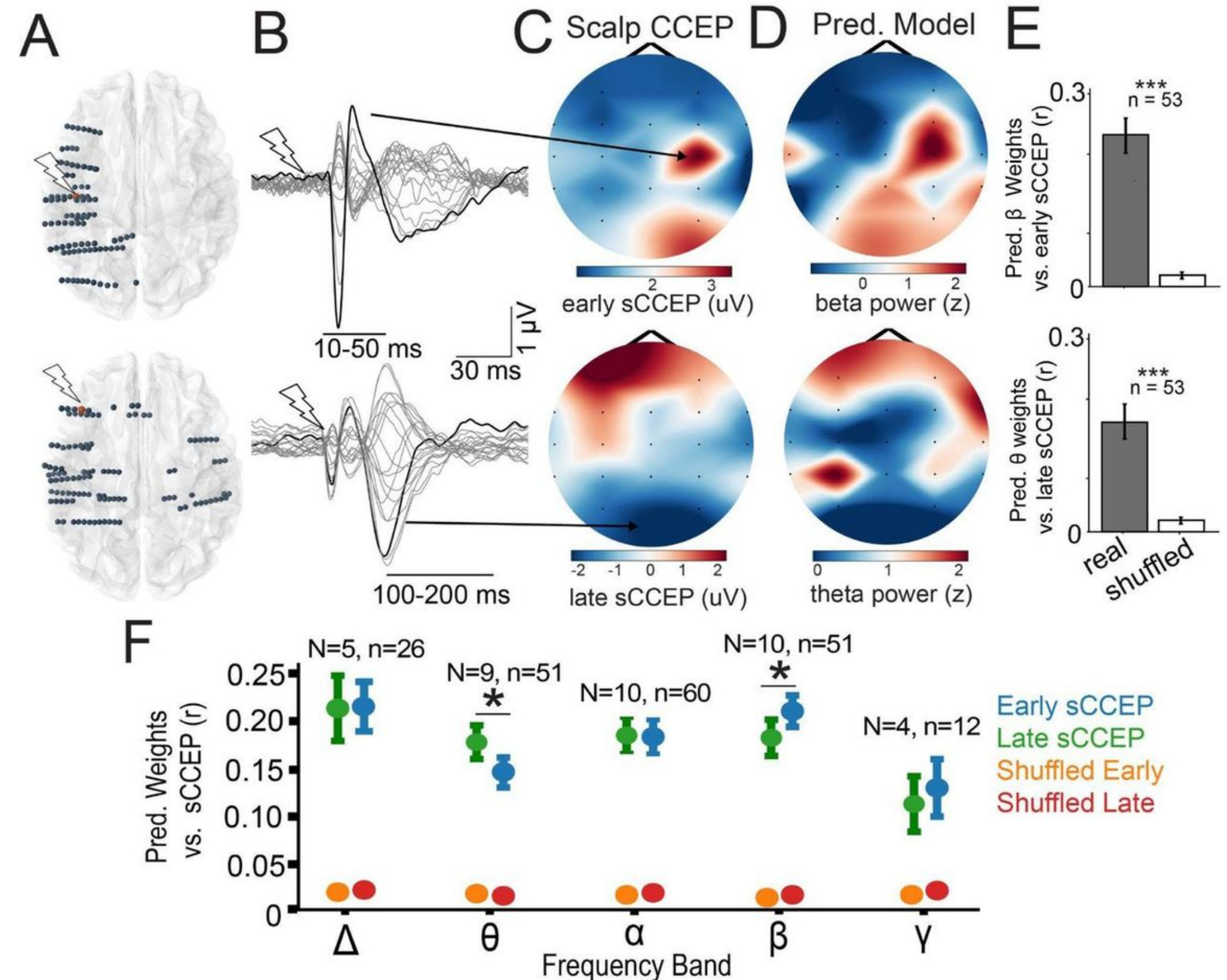
Current travels through CSF, resistive skull, and scalp, spreading laterally.

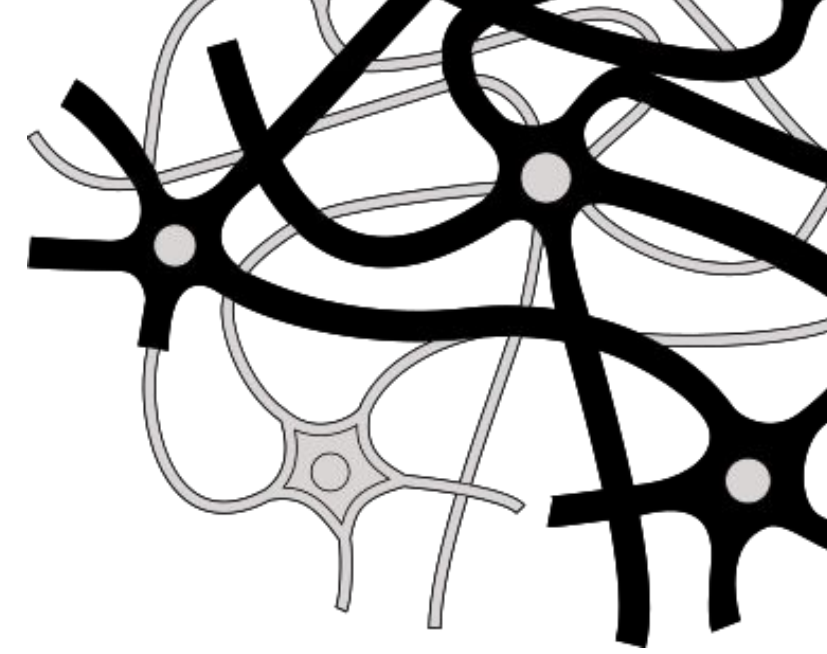
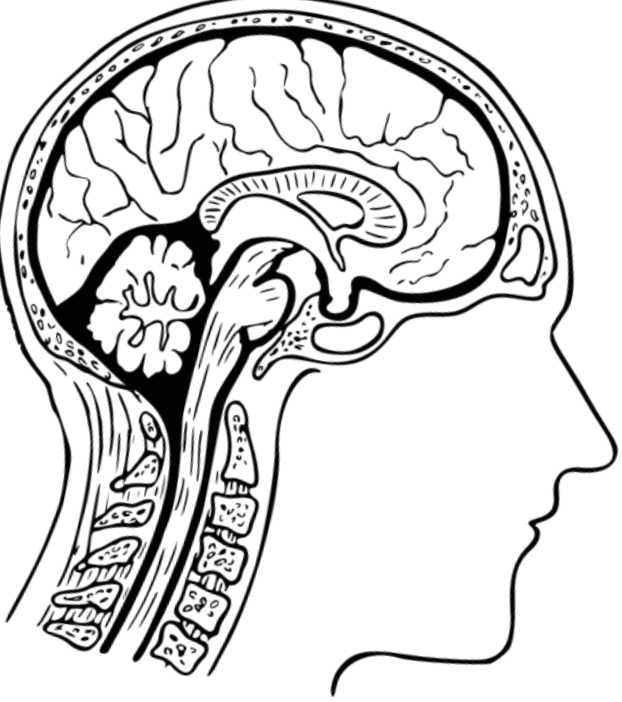
Forward Problem

Predicting scalp voltage from known brain sources (Mathematically solved).

Inverse Problem

Inferring sources from scalp data (Mathematically ill-posed).

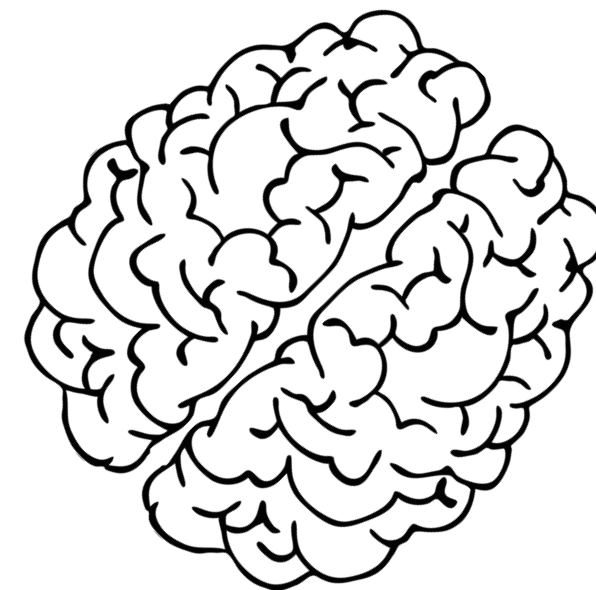
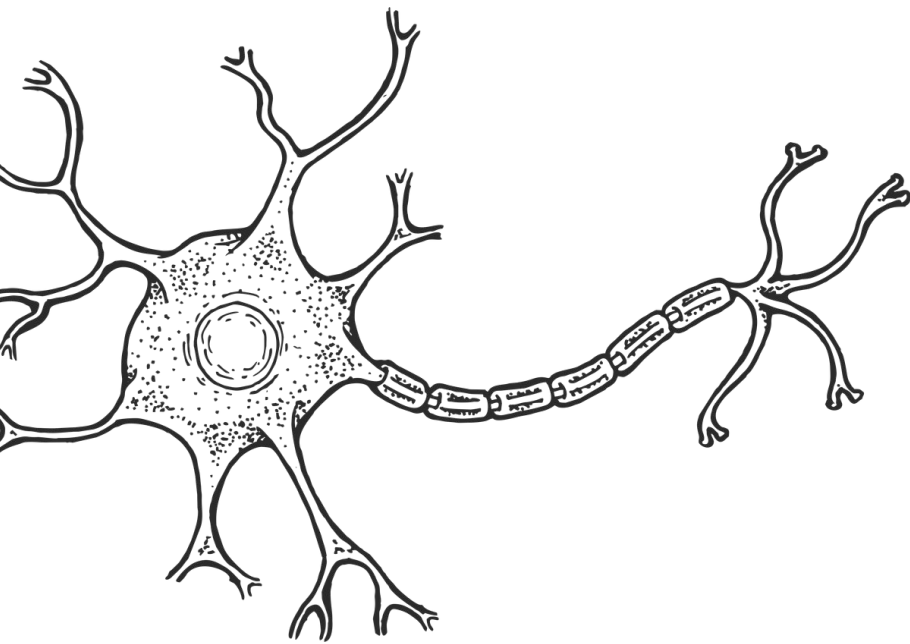


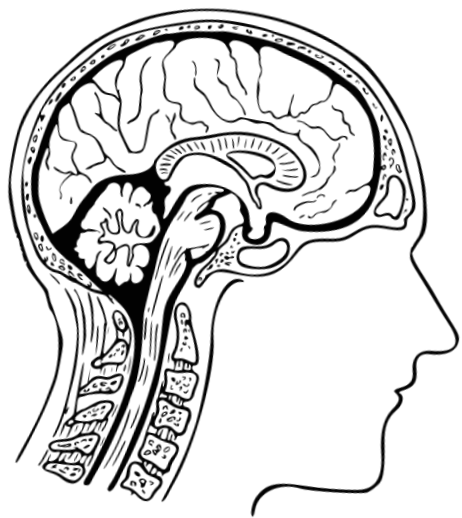


Session 2

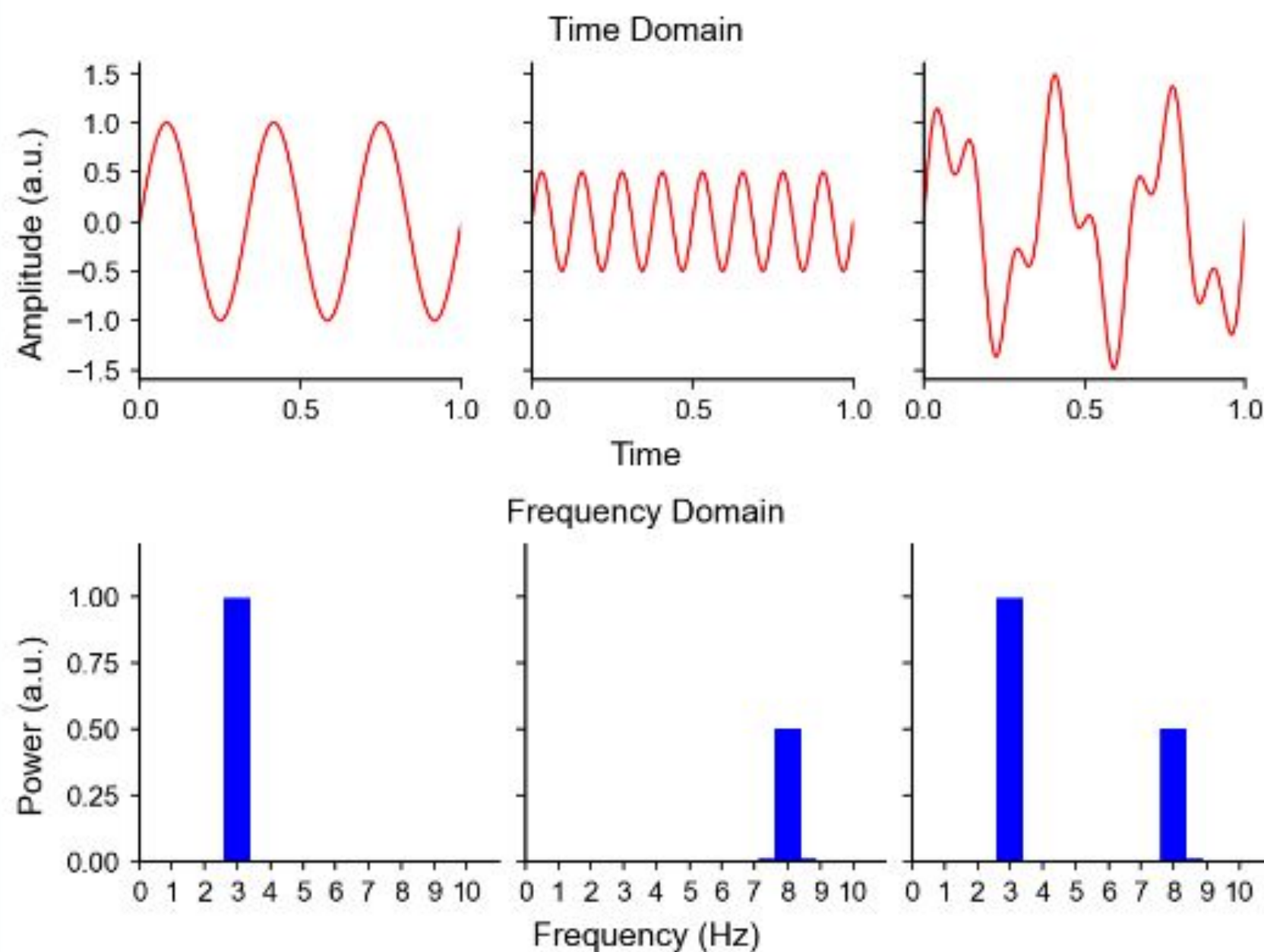
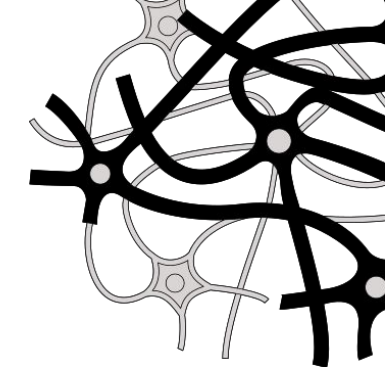
Neural Oscillations and Brain Rhythms

Decoding the Language of Frequency





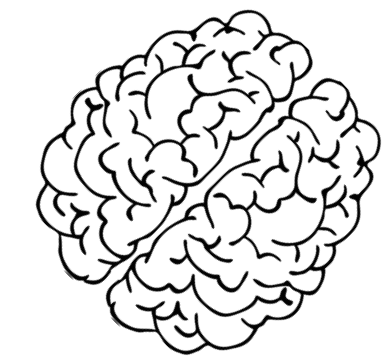
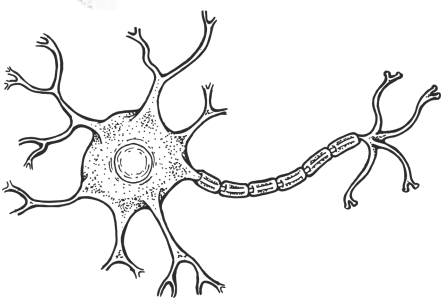
Fourier Perspective

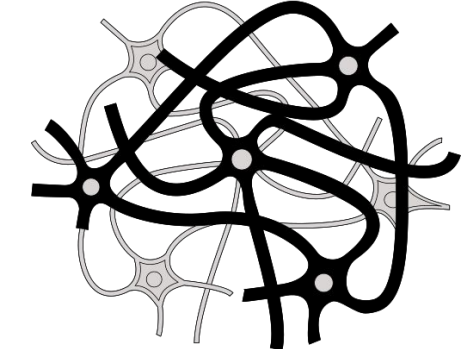
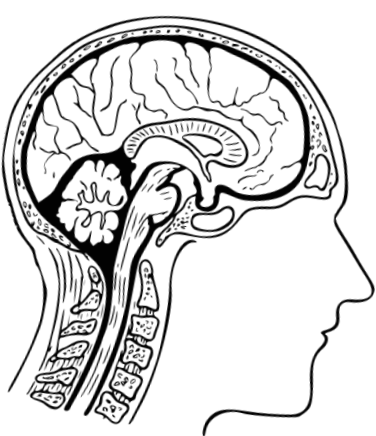


TIME DOMAIN VS FREQUENCY

Raw EEG is a complex, noisy time series. Any signal can be decomposed into constituent sine waves.

- » Time Domain: Voltage over Time
- » Fourier Transform: Mathematical Bridge
- » Power Spectrum: Energy at specific frequencies(Hz)

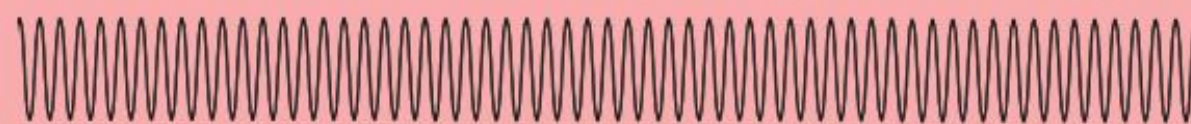




Frequency Dynamics

Gamma Waves

30 - 100 HZ



INSIGHT
PEAK EXPERIENCES
SYNCHRONIZATION

Beta Waves

12 - 30 HZ



ALERTNESS
CONCENTRATION
THINKING

Alpha Waves

8 - 12 HZ



MEDITATION
CREATIVITY
RELAXATION

Theta Waves

4 - 8 HZ



VISUALIZATION
TRANCE
DREAMING

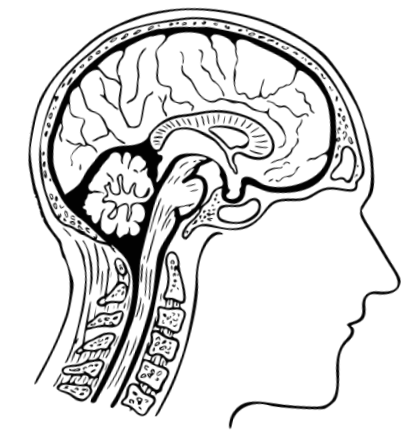
Delta Waves

0.5 - 4 HZ

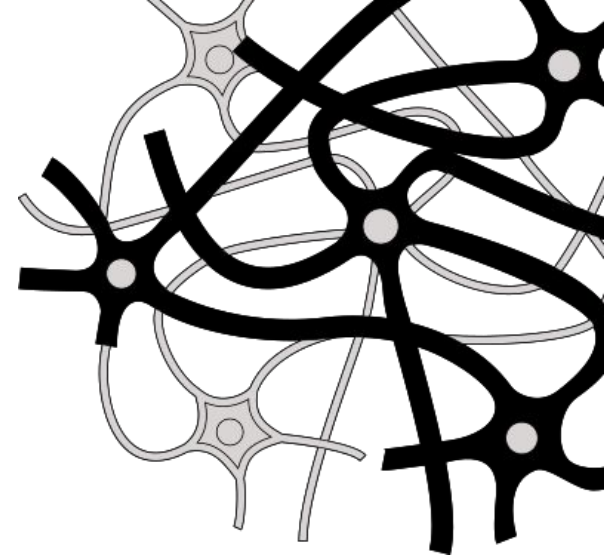


DEEP SLEEP
TRANSCENDENCE
RESTORATION

- **Delta (-4Hz):** Slow-wave sleep, structural pathology.
- **Theta (-8Hz):** Memory encoding, hippocampal coordination.
- **Alpha (8 -13Hz):** Visual idle state (occipital dominance).
- **Beta (13 - 30Hz):** Active concentration, motor execution.
- **Gamma (30 - 100+Hz):** Sensory integration, local network computation.



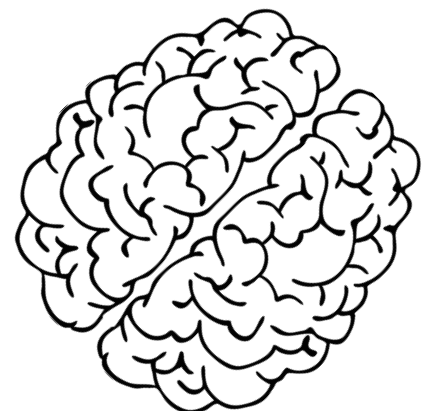
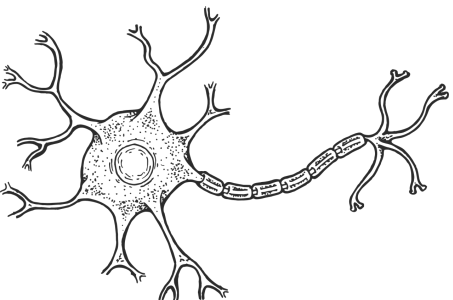
Improving Signal to Noise



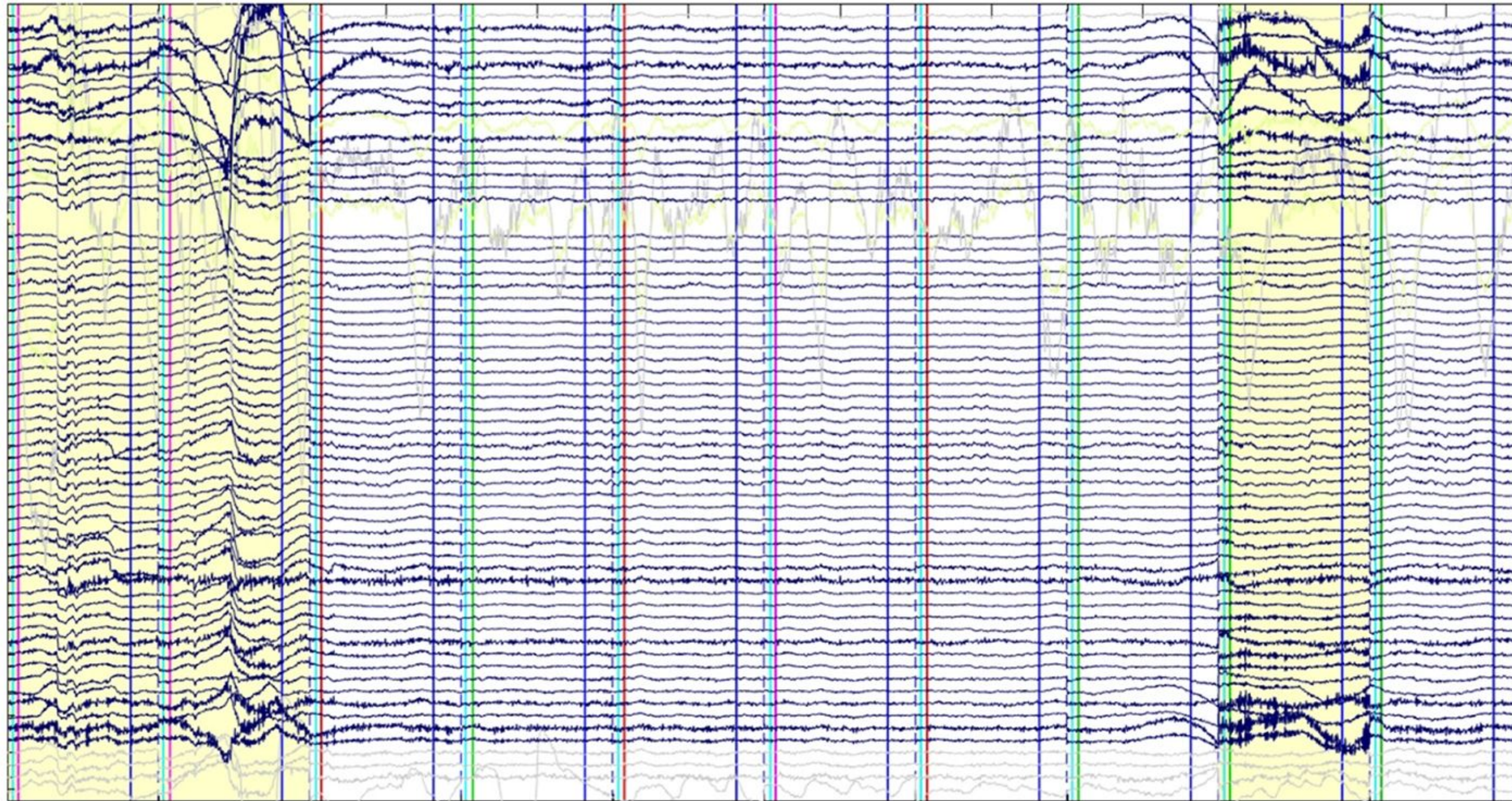
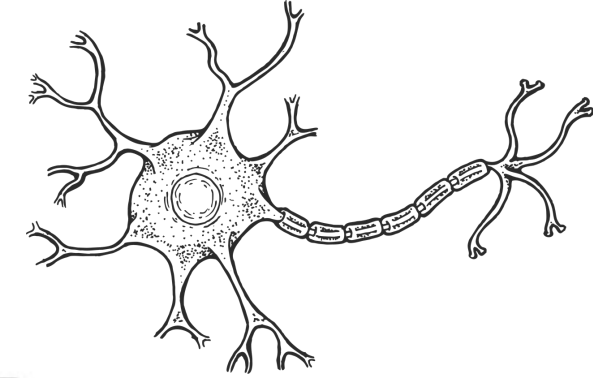
Event-related Potentials

- **The Challenge:** Brain responses are tiny ($\approx 5\mu\text{V}$) buried in noise ($\approx 50\mu\text{V}$).
- **The Solution:** Averaging. Consistent responses add up; random noise averages to zero
- **The maths:** $SNR \propto \sqrt{N}$
- **Tradeoff:** To double your signal quality, you should quadruple your trial count ($4N$)

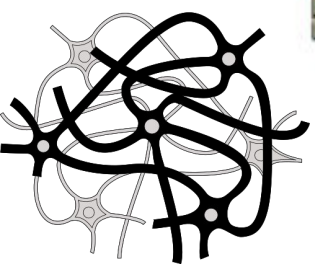
$$SNR \propto \sqrt{N}$$



Channel

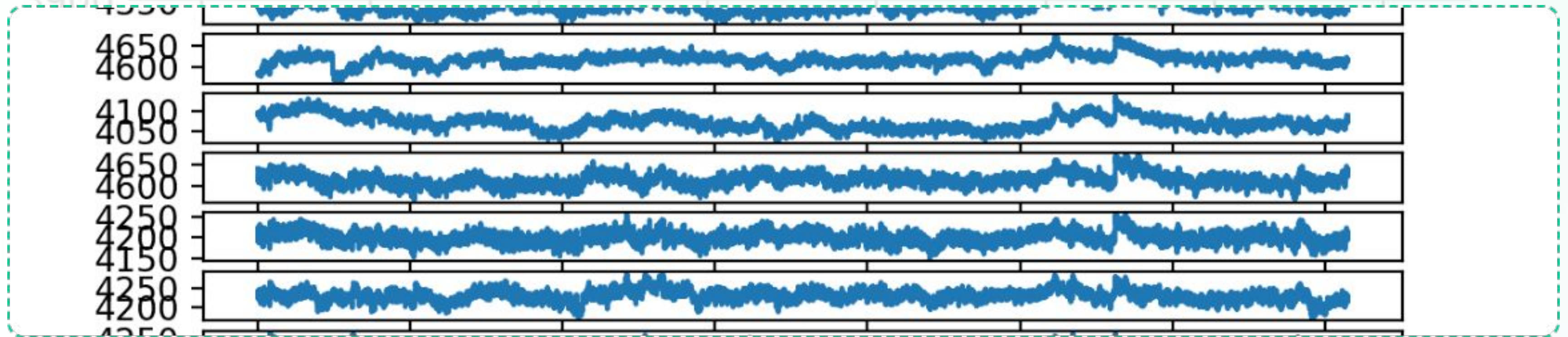


Trial (1-10)



Knowledge Integration Check

Analyze the visual trace below:



Prompt: Look at the 10-second trace. At second 4, the amplitude drops completely. Identify the frequency shift and the biological mechanism.

Answer: Alpha rhythm desynchronization (Alpha Blocking) due to eyes opening.

Image Sources



https://www.hopkinsmedicine.org/-/media/images/health/1_-conditions/brain/brain-main-parts.png

Source: www.hopkinsmedicine.org



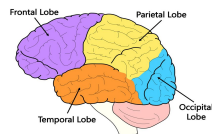
https://med.libretexts.org/@api/deki/files/52410/Brain_Coronal_Section_-_Grey_and_White_Matter.png?revision=1&size=bestfit&width=799&height=479

Source: med.libretexts.org



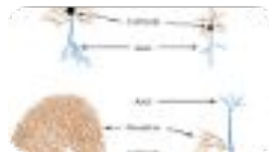
<https://webvision.pitt.edu/wp-content/uploads/2018/05/fig16cortex3d.jpg>

Source: www.webvision.pitt.edu



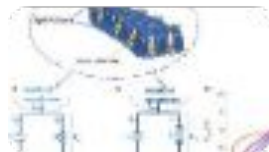
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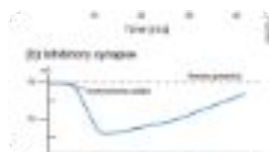
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Source: openpress.sussex.ac.uk



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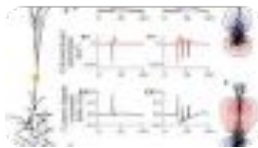
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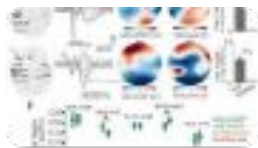
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Image Sources



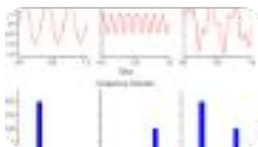
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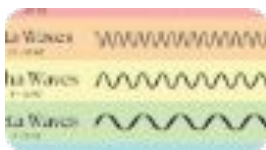
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Source: dibsmethodsmeetings.github.io



https://cdn.shopify.com/s/files/1/0348/7053/files/storage.googleapis.com-486681944373284_61cb9936-f6c2-493d-8402-3426d7f5a049_1024x1024.jpg?v=1689309340

Source: choosemuse.com



<https://www.diygenius.com/wp-content/uploads/2022/12/brainwave-frequencies-chart.jpg>

Source: www.diygenius.com



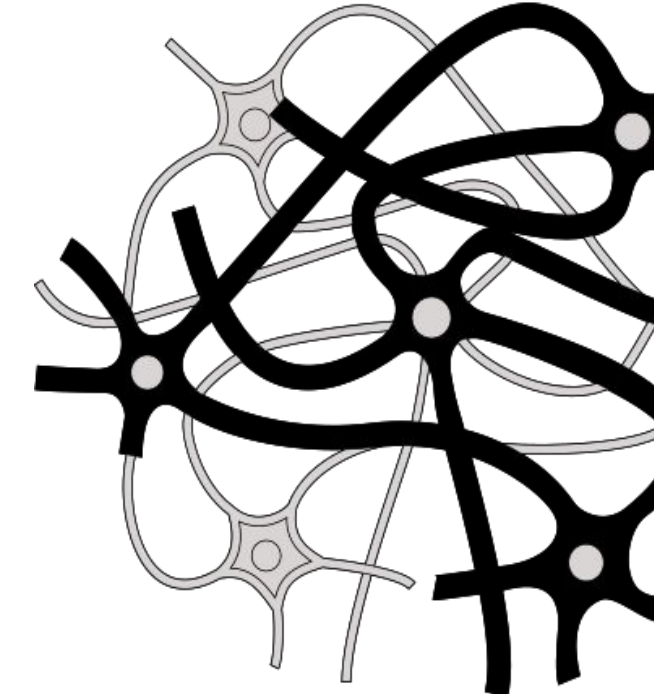
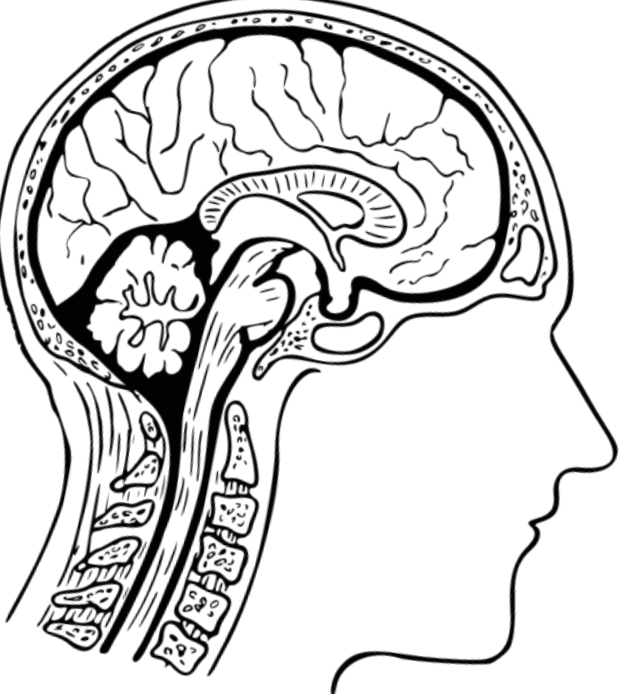
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<https://machinelearningmastery.com/wp-content/uploads/2018/06/Line-Plot-for-each-EEG-trace-and-the-output-variable-without-outliers.png>

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**Thank
You**

